

AI adoption, productivity and employment: Evidence from European firms

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The mission of the EIB Economics Department is to provide economic analyses and studies to support the Bank in its operations and in the definition of its positioning, strategy and policy. The department and its team of economists is headed by Debora Revoltella, director of economics.

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Abstract

This paper provides new evidence on how the adoption of artificial intelligence (AI) affects productivity and employment in Europe. Using matched EIBIS-ORBIS data on more than 12,000 non-financial firms in the European Union (EU) and United States (US), we instrument the adoption of AI by EU firms by assigning the adoption rates of US peers to isolate exogenous technological exposure. Our results show that AI adoption increases the level of labor productivity by 4%. Productivity gains are due to capital deepening, as we find no adverse effects on firm-level employment. This suggests that AI increases worker output rather than replacing labor in the short run, though longer-term effects remain uncertain. However, productivity benefits of AI adoption are unevenly distributed and concentrate in medium and large firms. Moreover, AI-adopting firms are more innovative and their workers earn higher wages. Our analysis also highlights the critical role of complementary investments in software and data or workforce training to fully unlock the productivity gains of AI adoption.

JEL Codes: D22, J24, L25, O33, O47

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1 Introduction

Artificial intelligence (AI) is poised to transform the global economy by simultaneously affecting aggregate demand and productivity on the supply side. Yet robust evidence on its firm-level effects outside the United States (US) remains scarce. While optimistic projections predict that AI could drive productivity booms (Baily, Brynjolfsson, and Korinek, 2023), others caution that adoption barriers and productivity constraints from unbalanced growth may significantly limit the gains from AI adoption (Acemoglu, 2024; Filippucci, Gal, and Schief, 2024). Europe’s position is both central and paradoxical: it boasts world-leading researchers and strong industrial capacity but suffers from a persistent and growing gap in the development of new AI technologies compared to the US and China (EIB, 2025). Understanding how AI affects European firms is therefore essential for policymakers seeking to harness its benefits while managing the associated risks.

This paper provides the first causal evidence on how AI adoption affects productivity and employment in European firms, leveraging a unique survey on EU and US non-financial firms matched to administrative financial data. We address two core questions. First, we explore the firm characteristics that predict AI adoption in Europe, which is key to characterize the selection process into using AI tools. We then investigate how AI adoption influences firm productivity and employment. The first question thus provides the empirical groundwork for a credible identification of the second.

The analysis reveals four key insights. First, AI adoption increases labor productivity by 4% on average after addressing endogeneity. Second, the gains stem from capital deepening rather than job displacement. Third, benefits are concentrated in medium and large firms – posing risks of widening inequality in the benefits from AI adoption. Fourth, the productivity gains are larger if firms carry out complementary investments in software and data or employee training. Our findings cast some doubts on claims of dramatic AI-driven job destruction, while underscoring pressing concerns about its uneven economic impact.

We leverage data of the European Investment Bank Investment Survey (EIBIS), an annual survey covering approximately 12,000 non-financial firms across the EU and 800 US non-financial firms, matched with ORBIS data on corporate financial statements. Our final sample used in the analysis is a pooled cross-section that runs from 2019 to 2024. The survey’s stratified design ensures representative coverage across firm size, sector and

country ([Brutscher et al., 2020](#)), while its detailed questions identify AI adoption based on whether firms use big data analytics and AI technologies in their operations. Crucially, EIBIS also captures granular firm characteristics on investment behavior, investment finance and innovation activity, enabling a rigorous analysis of both the drivers and consequences of AI adoption.

Three stylized facts emerge from the data, informing our approach for causal identification of the effects of AI. First, adoption is highly stratified: 45% of large firms (more than 250 employees) deploy AI, compared to only 24% of small firms (10 to 49 employees).¹ This echoes technology diffusion theories ([Comin and Hobijn, 2010](#)). Second, emerging EU economies (e.g., Romania, 22%) exhibit adoption rates substantially below more financially developed EU countries (e.g., Sweden, 52%) and the US (34%). Third, AI-adopting firms are more innovative in general, invest more intensively, and face tighter skilled-labor constraints – suggesting endogenous adoption patterns that require robust identification.

To establish causality, we develop a novel instrumental variable (IV) strategy inspired by the work on financial dependence and growth following [Rajan and Zingales \(1998\)](#). We match each EU firm with comparable US firms (same sector and size) that exhibit similar levels of innovation intensity, investment, managerial practices and external finance. We then assign the US firms’ AI adoption rate as an exogenous proxy for EU firms’ adoption rate. This approach leverages cross-continental variation in AI exposure, while holding firm characteristics constant, effectively isolating the effects of AI adoption. Propensity-score balancing tests confirm the validity of the identification strategy, with matched US and EU firms exhibiting near-identical distributions across key firm characteristics.

Our results show a robust causal link between AI adoption and productivity. Controlling for a wide set of country, sector and year fixed effects and observable financial variables, we find that firms adopting AI have 4% higher labor productivity. This effect is economically significant and aligns with mid-range macroeconomic projections recently reported in the literature ([Acemoglu, 2024](#); [Bergeaud, 2024](#)), rather than with optimistic “productivity boom” scenarios ([Baily, Brynjolfsson, and Korinek, 2023](#)). Crucially, the gains reflect capital deepening: AI augments worker output without reducing employment, consistent with micro-level evidence of AI-aided efficiency gains in cognitive tasks ([Noy and Zhang, 2023](#); [Gambacorta et al., 2024](#)). Moreover, we find that workers in

¹The shares are calculated using value added (VA) weights.